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Nigeria Road Management System: A Citizen-Centric Platform for Road Monitoring, Reporting, and Progress Tracking

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ABSTRACT:

KEYWORDS: citizen reporting, construction progress tracking, crowdsourcing, inspection alerts, multi-tier architecture, Nigeria roads, spatial dashboard, transparency.

I. INTRODUCTION

Road transport is the main mode of transport for both passengers and goods in Nigeria. The national road network that covers an estimated 193,200 kilometres is the largest in West Africa (Farouq *et al.*, 2017). This network is the main roadway of the cities, peri-urban areas and rural regions and accounts for more than 90% of passenger and goods travel. Despite its strategic significance, a significant part of the network is still in disrepair. The common sightings by motorists and pedestrians are deep potholes, structurally unsafe bridges, eroded shoulders and flooded carriageways (Adepoju, 2021). This chronic degradation causes a progressive deterioration of the vehicle, long delays and higher frequencies of accidents, which puts a significant socio-economic burden on society and the economy (Jaiye & Elisha, 2020).

Institutional fragmentation is a feature of the governance of road infrastructure in Nigeria. Federal roads are managed by the Federal Ministry of Works in consultation with the Federal Road Safety Corps (FERMA), State roads are managed by State Ministries of Works and Local roads are managed by Local Government Councils (Ogbu & Ezeh, 2019; Olojede *et al.*, 2019). The Traffic law enforcement is delegated to different State traffic agencies (STAs) and the Federal Road Safety Corps (FRSC) (Ajiboye *et al.*, 2020). In practice, such institutions are highly reliant on manual inspection, telephonic complaints, and paper documentation (Odinfono & Olawole, 2024). Unfortunately, in many cases of degraded roads, the road does not get logged for a long time before it's considered for a remedy. As such, common traffic violations (such as ignoring one-way signs, speeding, parking on private property, and littering in drains) are often unenforceable because of the lack of a citizen-friendly evidence reporting system (Umar *et al.*, 2021).

Another important challenge is the lack of transparency in the construction and rehabilitation of the roads. Inadequate materials have been used by contractors and some contractors have been seen either leaving project sites after mobilisation funds have been released or leaving them before any subsequent stages of the project have been completed (Adepoju, 2021; Ugochi, 2020). Since there is no clear digital record of progress using photographs, it's very hard for supervisory personnel to see if contractual requirements have been met. Some of these can be directly addressed through a computerised system that allows citizens to report road defects and traffic violations, requires the contractors to submit progress photos and automatically alerts when the deadlines for inspections have passed.

The current road management practices in Nigeria are mostly manual, inefficient and uncoordinated. This prevailing system has four major drawbacks. First, citizens have no efficient digital system to submit their complaints about potholes or structural damage to roads with a geotagged photo, and complaints can only be made at the end of a physical visit or informally, which means that the repair process takes too long (Farouq *et al.*, 2017). Secondly, construction activities are not recorded digitally, using an open format with photographic evidence, which allows for inferior construction methods and missed deadlines to remain unchecked (Adepoju, 2021). Third, traffic violations like one-way contraventions, illegal parking and dropping of waste on roads occurs without any form of regulation in place, without citizens having any way to submit supporting photos with evidence (Umar *et al.*, 2018). Fourth, maintenance agencies do not have an automated notification system that can alert when a road or project inspection is due, resulting in poorly maintained infrastructure and maintenance schedules (Odinfono & Olawole, 2024).

To meet the needs of these exigencies, this study addresses three main research questions. Are there any existing systems for the management of road construction, maintenance, monitoring and enforcement of road traffic laws in Nigeria? What are the main drawbacks of current methods of road management, in terms of transparency, coordination between agencies, material quality, and public reporting? What do you think the functional attributes of a computerised Nigeria Road Management System should include to enable citizen reporting, track the progress of road construction, monitor the state of the roads and improve their decision-making by officials?

This study intends to develop, deploy and test a web-based Nigeria Road Management System (NRMS) that facilitates citizens to report defects and violations, monitor the construction progress and notify officials of outstanding inspections.

The study addresses the aim through three specific objectives: The first goal is to design the multi-level architecture of the system, the database schema, and the UML design for contractors, road officials and users. The second goal is to implement the core software modules, including citizen reporting of defects and violations with photographic upload, contractor progress monitoring, automated alerts for inspections and an administrative monitoring dashboard. The third goal is to test the final system to ensure a successful functioning, reporting and notification in representative scenarios.

II. RELATED WORKS

The results of research on road management and transport technology in Nigeria can be categorized into three main groups: (i) administrative and institutional record-management systems, (ii) geospatial and emergency-response systems, and (iii) automated sensing and computer-vision systems. While these studies show the possibilities of the use of digital tools, they are institution-centered, limited and mostly isolated from public inclusion. It synthesises all the literature which has been audited with respect to the five variables: system architecture, core features, technology stack, problem solved, and unaddressed gaps and finally identifies the gap which the proposed Nigeria Road Management System (NRMS) addresses.

2.1 Administrative and Institutional Record-Management Systems

One group of research is dedicated to the digitisation of the records of the government itself. The Federal Road Safety Corps (FRSC) faced problems of traffic offence information being decentralised in states and local government areas (LGAs) which Sese (2020) addressed through a client–server distributed database. The solution was to solve a real issue of record fragmentation for internal use, but had no citizen facing reporting, photo uploads or road condition tracking. In a similar manner, Dabo and Yunus (2020) showed that GIS tools can be used to map the road facilities in Katsina, however, the GIS system was limited to an internal database and was not accessible to the common man. Akanbi and Abegunde (2020) extended this institutional logic to a mobile solution for vehicle plate registration and accident logging for the FRSC which lacked the reporting of road-surface defects and monitoring of construction progress.

The overall architecture principle that underlies these systems is that the citizen is not the subject of data, but rather is purely passive. Information is moving towards the agency, and there is no flow of information back to the public. This will constrain accountability and the amount of road condition data that can be gathered at ground level. Farouq et al. (2017) provide empirical evidence on the impact of this limitation: they found severe issues with the manual, paper-based reporting system, and citizen discontent about the process of reporting road-defects in Kano State. Their study did not, however, result in a working web or mobile platform.

2.2 Geospatial and Emergency-Response Systems

A second cluster is related to the use of spatial analysis in road safety. Audu et al (2021) created a spatial decision support system (SDSS) with ArcGIS 10.3, Network Analyst and GPS to map accident hotspots and to calculate the shortest ambulance routes in Ibadan. The system enhanced the response times for all emergencies and showcased the benefit of geospatial intelligence to road management. However, its coverage was strictly post-crash: It only covered emergency medical response, and did not include anything like defect reporting, preventive maintenance or contractor tracking. The system thus addresses a downstream symptom and not an upstream problem, of poor road quality.

2.3 Automated Sensing and Computer-Vision Systems

The third cluster delves into automated enforcement and monitoring. Umar et al. (2021) developed an image processing client–server system to detect vehicles running the red light that uses licence plate detection and optical character recognition (OCR) to identify the licence plate of the vehicles using k-NN algorithm and MATLAB. The system enabled automatic detection of violation, but it needed to be based on a specialised fixed infrastructure and had no functionality for citizen reporting and road-condition monitoring. It is capital intensive and location dependent, and is hard to deploy to spread out and low-cost road monitoring requirements in Nigeria.

2.4 Cross-Cutting Limitations and Data-Quality Concerns

There are a number of cross-cutting limitations which are identified across all three clusters. There is a systematic lack of involvement of the public. While Toriola-Coker et al (2020) found that citizens' lack of involvement in public roadworks was a major issue that leads to the failure of public road projects, none of the systems audited provides a way for citizens to report defects and/or verify progress of the contractor. Secondly, there is an inconsistent data collection. According to Odinfono and Olawole (2024), 55% of the traffic crash data collection is not done using standardized paper forms in southwestern Nigeria, resulting in inconsistencies between police and road-safety records. Third, they are not integrated systems, as each system solves a small part of the road-management problem, without the citizen crowdsourcing, contractor milestone tracking and inspection alert systems all being in one place.

2.5 Summary of Audited Systems

TABLE I: Summary of Related Systems Audited Across the Five Project Variables

Study & Year	Architecture	Core Features	Tech Stack	Problem Solved	Unaddressed Gap
Afieroho *et al.* (2023)	Web E-Government Framework	Public project tracking, citizen adoption survey	SPSS, PLS-SEM, Web Portal	High public willingness to adopt digital tracking tools	Theoretical survey model only; no functioning tool with photo reporting

Farouq *et al.* (2017)	Relational Database Audit	Road defect logging, user dissatisfaction survey (Kano)	SPSS, Average Index Analysis	Exposed weaknesses and frustration with manual defect logging	Fully manual and paper-based; no web or mobile platform
Sese (2020)	Client–Server Distributed DB	Centralised traffic offence database for FRSC	Client–Server, Distributed SQL	Centralised offence records across states and LGAs	Internal officers only; no citizen reporting, photo uploads, or road tracking
Audu *et al.* (2021)	Spatial Decision Support (GIS)	Accident hotspot mapping, ambulance routing (Ibadan)	ArcGIS 10.3, Network Analyst, GPS	Improved emergency response times	Post-crash only; excludes defect reporting and contractor tracking
Umar *et al.* (2021)	Image Processing Client–Server	Licence plate detection and OCR	k-NN, MATLAB, Camera	Automatic detection of red-light violations	Requires fixed roadside cameras; no citizen reporting or condition monitoring
Dabo and Yunus (2020)	GIS Internal Database	Mapping of road facilities (Katsina)	GIS	Effective mapping of road assets	Internal database; inaccessible to ordinary citizens

Akanbi and Abegunde (2020)	Mobile System	Vehicle plate registration, accident logging (FRSC)	Mobile Platform	Digitised plate registration and accident logs	No road-surface defect reporting or construction-progress monitoring
Odinfono and Olawole (2024)	Survey / Data Audit	Analysis of crash data collection practices	Survey Analysis	Revealed 55% reliance on non-standard paper forms	No system proposed; diagnostic only
Toriola-Coker *et al.* (2020)	Conceptual / Empirical Study	Public participation in road projects	—	Identified lack of public involvement as failure factor	No digital platform proposed

2.6 Proposed Improvement and Value-Add Delta

The literature available for review shows that the current systems are institutional based, function separately and do not involve the public. The proposed Nigeria Road Management System (NRMS) combines citizen crowdsourcing, contractor milestone tracking and automated alerts for inspections within a single platform which addresses these gaps. Table II is a comparison of standard existing practice to the proposed contribution for NRMS.

TABLE II: Value-Add Comparison (Standard Practice vs. Proposed NRMS)

Functional Area	Standard / Existing Practice	Proposed NRMS Contribution (Value-Add Delta)
Citizen Defect Reporting	Slow verbal complaints, physical office visits, or unverified social media posts without precise location data	Web and mobile interface for reporting potholes and violations with attached photos and automatic GPS coordinates

Contractor Progress Tracking	Contractor-submitted paperwork with little or no public photo verification before payment	Dedicated contractor portal requiring timestamped progress photos for each construction milestone before inspection clearance
Inspection Alerts	Irregular manual visits with no automated reminder when inspections are overdue	Automated background engine that checks the database daily and sends email/SMS alerts when a road inspection is overdue (> 30 days)
Official Dashboard & Analytics	Disorganised paper files and separate spreadsheets with no interactive map or condition status	Central dashboard showing monitored roads, critical road counts, ongoing construction projects, and overdue inspections on an interactive map

The value add delta is thus not simply additive but architectural, as NRMS transforms the passive citizen as a subject and turns institutional records into a platform of public accountability and turns manual inspection with rules into automated monitoring.

III. METHODOLOGY

3.1 Research Design

The methodology used in this study is Design Science Research (DSR) that is suitable for research where the significant contribution is an innovative software artefact. As part of information systems and software engineering, DSR is a paradigm that is well accepted as one that focuses on the design and evaluation of artefacts for solving real-world problems, as explained by Hevner et al. in 2004. Design science differs from behavioral science, which aims to explain or predict phenomena, by producing knowledge using design and evaluation of artefacts, as pointed out by Hevner et al. in 2004.

The DSR process is based on the design cycle developed by Wieringa (2014), which involves two activities: designing an artefact that will enhance an aspect of the context for stakeholders and then empirically testing the artefact in the context to see how well it works. This “validation in context” principle is at the heart of the methodology: Not only is an artefact created but it must also be demonstrated as solving the identified problem in the operational context, a point made by Wieringa in 2014.

The architecture of the system is fully documented before the implementation starts, and the system requirements are traceable to the changes made in its design.

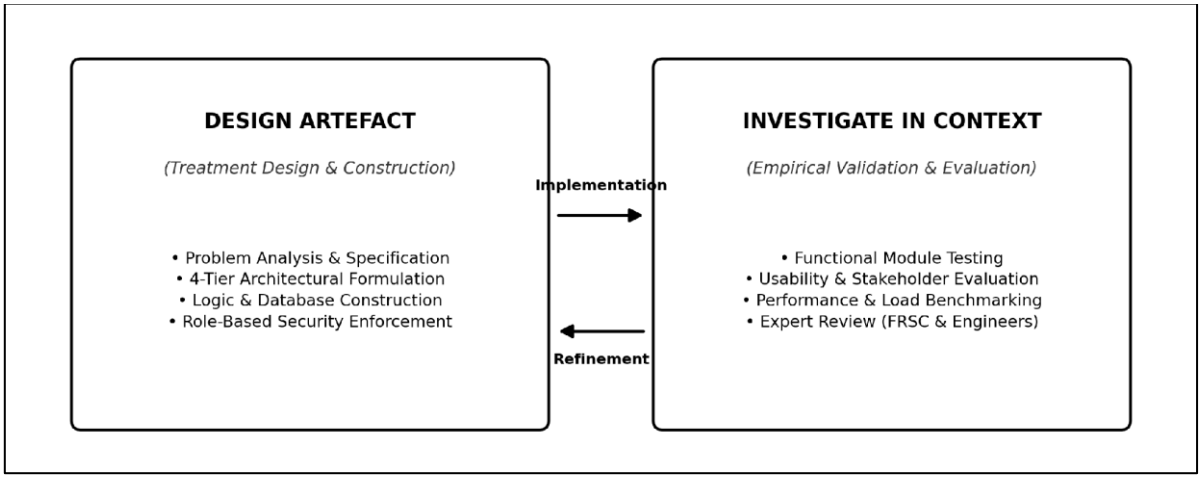


Figure 3.1: Wieringa's (2014) Design Science Research Cycle Applied to NRMS.

3.2 System Architecture

The Nigeria Road Management System (NRMS) is based on a four-tier architecture for separation of concerns, which includes presentation, business logic, data persistence and notification services.

TABLE III: Architectural Tiers and Component Allocations

Tier	Responsibility	Components
Client Tier	User interaction	Citizen Portal, Contractor Portal, Officials Dashboard
Logic Tier	Business rules and workflow	User Management, Report Management, Contractor Progress, Inspection Alert Engine, Map Analytics
Data Tier	Persistence and caching	PostgreSQL + PostGIS, Cloudinary/S3, Redis
Output Tier	Notifications and reporting	SMS/Email Services, Map Rendering (Leaflet/OpenStreetMap), PDF/CSV Export

This separation provides three design properties: security (role-based access is enforced on the server), maintainability (an independent modification of tiers, without redesign of clients), and scalability (horizontal scaling without redesign of clients).

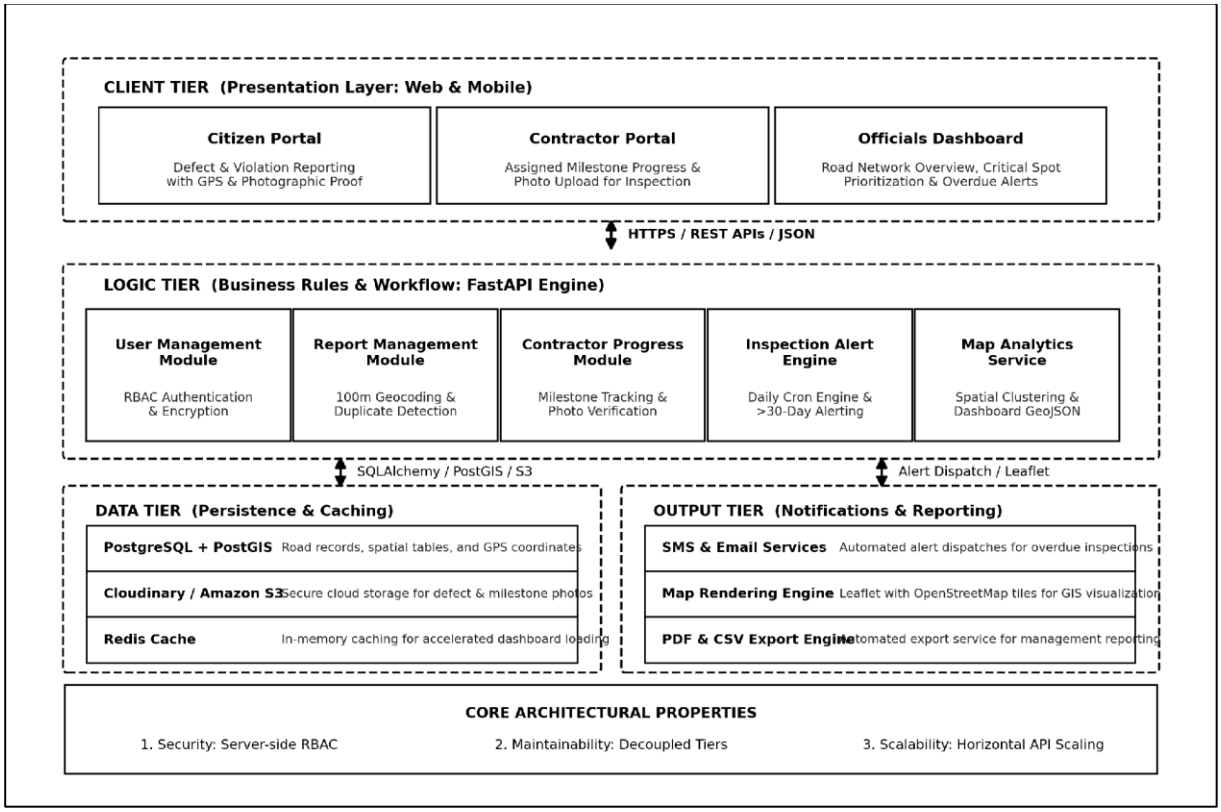


Figure 3.2: Detailed NRMS Four-Tier System Architecture.

3.3 Functional Modules

Client Tier. A Citizen Portal allows users to submit road defects (such as potholes and erosion) and traffic violations (such as driving on one way and parking illegally) through an instant web and mobile interface, capturing photos and GPS coordinates. The Contractor Portal allows registered contractors to view the projects assigned to them, upload of the progress pictures with date and time stamp for the completed milestones and request inspection. The Officials Dashboard will show road counts, critical road spots, current projects and overdue inspection counts.

Logic Tier. The User Management Module encompasses all the security measures required for registration of new users, the secure authentication of all users, the encryption of their passwords and role-based access control for four roles: citizen, contractor, inspector, and administrator. Citizen submissions are received by the Report Management Module, a 100-metre radius is used to detect if there is a duplicate report and an urgency priority is assigned. The Contractor Progress Module documents project progress, calculates work completed as a percentage and stores progress photos to verify progress. The Inspection Alert Engine runs on a daily basis to alert road sections or projects with an inspection date more than 30 days ago. The Map Analytics Service takes the GPS data and clusters reported defects geographically to show them together and formats the data to be displayed in a dashboard.

Data Tier. Road records and GPSCoordinates use the PostgreSQL/PostGIS spatial data storage database. Photographs are stored in Cloudinary/S3. Redis is used for caching on dashboards.

Output Tier. Notification Services send out SMS and email alerts for late inspections. Map Rendering is Leaflet + OpenStreetMap tiles. Report Generation is a tool that generates PDF/CSV reports to pass to management.

3.4 Data Flows

Four standard data flows (F1-F4) specify the interactions between actors, application logic and persistence layers during operation.

TABLE IV: System Operational Data Flows (F1 to F4)

Flow	Actor	Process	System Response
F1: Report Submission	Citizen	Selects defect type, attaches photo, submits GPS coordinates	Stores photo in cloud, persists report in database, returns tracking number
F2: Progress Upload	Contractor	Selects milestone, uploads progress photos	Records update as "Pending Inspection," notifies supervising officer
F3: Inspection Alert	System	Daily scan for inspections >30 days overdue	Generates alert, dispatches SMS/email to maintenance engineer
F4: Administrative Triage	Official	Reviews defects on map, assigns maintenance team	Status transitions from "Under Review" to "Assigned"

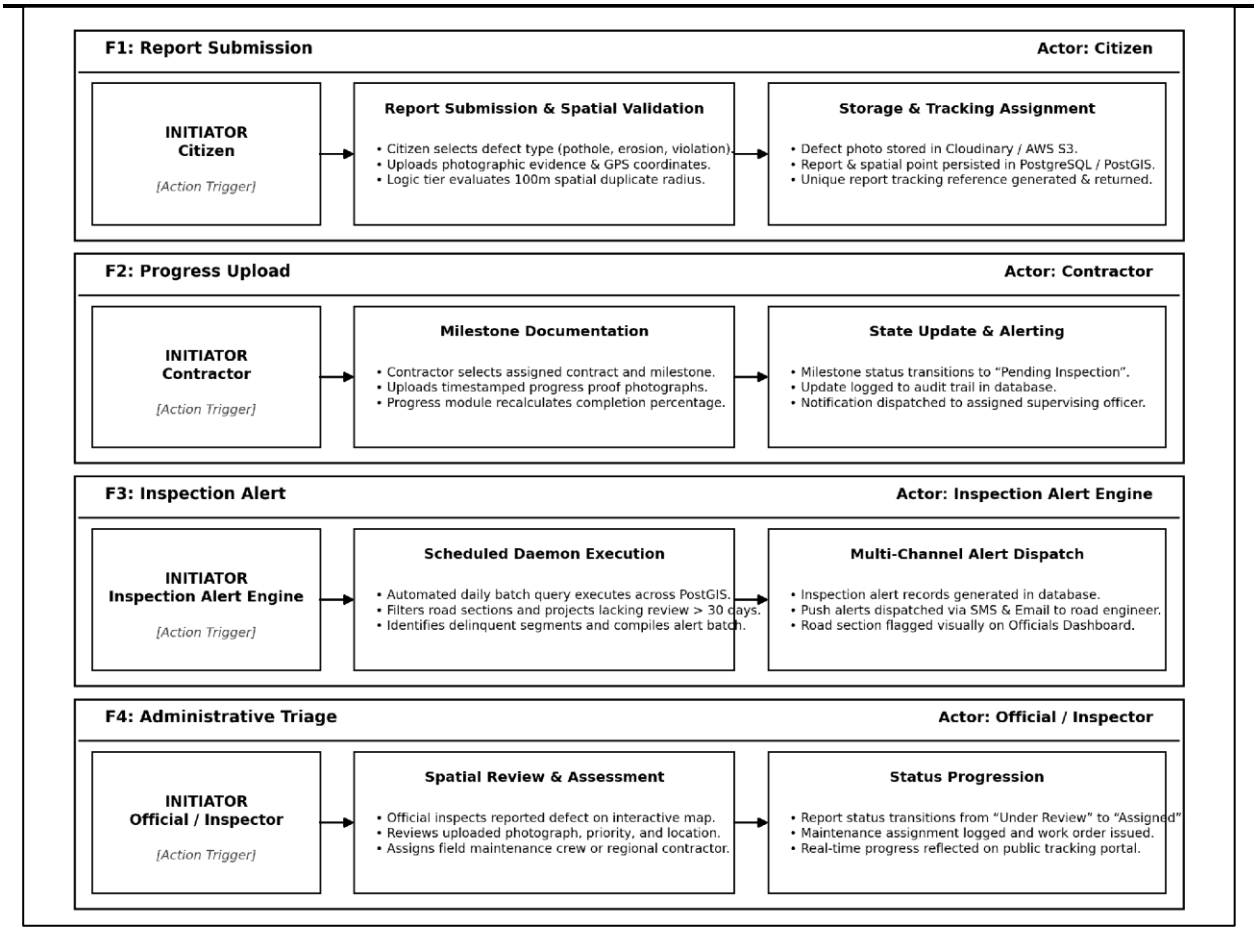


Figure 3.3: End-to-End System Operational Data Flows.

3.5 Technology Stack

The selection of technologies was based on three different criteria: reliability, open-source accessibility and ease of deployment.

TABLE V: Technology Selection and Rationale

Layer	Technology	Rationale
Frontend	HTML, CSS, JavaScript	Responsive mobile performance; dashboard and form support
Backend	FastAPI	High-throughput concurrent photo upload and API request handling
Database	PostgreSQL + PostGIS	Native spatial queries for GPS coordinates and geographic clustering
Storage/Cache	Cloudinary + Redis	Secure off-database media storage; accelerated dashboard loading
Mapping/Notification	OpenStreetMap + Leaflet; SMS/Email	Open-source interactive mapping; reliable alert delivery

3.6 Design Traceability

Architectural decisions are documented in relation to a gap in the literature. The 100-metre duplicate detection rule is for the issue of data quality that was highlighted by Odinfono and Olawole (2024) and concerns inconsistency in data. This is done by the automated 30-day inspection alert, which takes care of irregular manual inspection practices. The contractor photo-verification workflow is aimed at the lack of transparency in payments. This traceability allows the artefact to meet directly the deficiencies recorded and not just a technical exercise.

3.7 Evaluation Plan

The artefact will be assessed according to the four complementary assessment approaches mentioned in foundational design science literature and based on the DSR principles:

- Functional validation: Task each module – test them against the specified requirements - duplicate detection accuracy, alert trigger reliability.
- Usability evaluation: Task based testing using representative citizens, contractors and officials.
- User experience testing: API request times, photos upload rates and maps rendering times with simultaneous traffic.
- Expert Review: Structured walk-throughs with road engineers and FRSC officials to determine operational fit.

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